

Kate Ker

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Skills

Design Tools: Cadence Allegro, Altium, Soldering, SUE SoC Design Manager, MAX layout editor, SolidWorks, Rapid-Harness

Analysis Tools: Ansys HFSS, Keysight ADS, VNA, Oscilloscope, Logic Analyzer, CANoe, CANalyzer, LTSPICE, HSPICE, ModelSIM, Matlab

Programming: C, C++, Python, Verilog, Assembly, CAPL

Developer tools: CubeIDE, VSCode, Altera DE1-SoC, Git, Jama, Jira, Confluence

Experience

Tenstorrent - SI Jan. 2026 - Present
Signal Integrity Intern - Systems Engineering Toronto, ON

- Owned the full design life-cycle of three 112G PAM4 Ethernet daughterboards, executing pre-layout stackup design with vendors, schematic capture and layout using Cadence Allegro and Altium.
- Optimized Ethernet retimer and PCIe Gen5 channels and transitions using Ansys HFSS and Keysight ADS, evaluating S-parameters, eye diagrams, and BER.
- Designed coupons and using a 40GHz VNA, correlated real-world measurements of DUTs with simulations to refine future designs.

Tenstorrent - PCB Jan. 2026 - Present
Signal Integrity Intern - Systems Engineering Toronto, ON

- Owned the full design life-cycle of three 800G Ethernet daughterboards, executing schematic capture and HDI layout using Cadence Allegro and Altium, integrating I2C communication and low-speed control subsystems.
- Analyzed PCIe and Ethernet designs using Ansys HFSS and Keysight ADS to iteratively optimize layout for signal integrity performance, and verify channels met system requirements.
- Using a 40GHz VNA, correlated real-world measurements of DUTs with simulations to refine future designs.

Rivian Volkswagen Groups Technologies May 2025 - Dec. 2025
Electrical Controls Intern - Functional Safety Palo Alto, CA

- Conducted HARA and FMEA to define safety goals and derive functional and technical safety requirements for thermal, body, and infotainment systems, designing based on ISO 26262 standards.
- Designed and brought-up a UART-based, "man-in-the-middle" fault-injection tool to validate safety features. Then, expanded on this tool for functional validation in manufacturing facilities to accelerate production.
- Performed hardware test setup and debug for infotainment features, involving signal tracing and power measurements using oscilloscopes, logic analyzers and multimeters.

Rivian Automotive May 2024 - Aug. 2024
Electrical Engineering Intern - Functional Safety Irvine, CA

- Performed testing on Labcar/HIL system to validate Body and Thermal specific safety goals, ensuring compliance with functional safety standards.
- Assessed and executed 24 comprehensive, vehicle-level test cases tailored to technical safety requirements, ensuring all critical safety features were thoroughly evaluated.
- Created Python and CAPL test scripts to execute and automate hardware tests on different HIL setups.

University of Toronto Formula Racing Team Dec. 2022 - Aug. 2024
Low Voltage Architecture Lead Toronto, ON

- Led a multi-functional team of seven in the design and testing of the low-voltage vehicle power architecture, including power regulation and distribution, CANBus communication, and data acquisition systems.
- Designed, soldered, and debugged custom PCBAs, using oscilloscopes and multimeters for validation.
- Designed electrical enclosures on SolidWorks, from both sheet-metal and 3D print. Planned and manufactured harnesses architecture, reducing the weight by 60% from the previous year and improving reliability and manufacturability.

SED Lab, Faculty of Mechanical and Industrial Engineering May 2023 - Sept. 2024
Undergraduate Research Assistant Toronto, CA

- Compiled and synthesized findings on the effectiveness and future development of command and control centers. Recipient of the First Year Summer Research Fellowship from the University of Toronto.

Education

University of Toronto Sept. 2022 - May 2027
Electrical & Computer Engineering, Undergraduate

Projects

Health Tracker using STM32 and LoRa Module

Dec. 2025

- Designed and implemented an embedded health tracker in C using STM32 microcontroller, LoRa module (wireless), and integrated heart-rate and accelerometer sensors.

TX Power Amplifier and Filter for Flexible Radio Transceiver

Jan. 2023 - May 2023

- Applied RF design principles and coursework knowledge to design a PCB using Altium, then soldered and tested the board. Generated 1W of continuous power to a 50Ω load, from a 8-16MHz input signal.

HV Measurement Point PCB

Feb. 2023

- Designed a PCB using Altium, which acts as a safety-critical sensor. Uses using an optocoupler to determine measured voltage, and controls an LED indicator and relay to a discharge resistor.